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Improved Method for Enhancing Dark Images based on CLAHE and Morphological Reconstruction

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Abstract

The image acquired are often flawed due to various defects from environmental scenarios such as indoor lighting, cloudy weather etc especially at night time. The dark image contains compressed dynamic range that can be enhanced for understanding the detailed information. In this investigative work, an improved method for enhancing extremely dark images is proposed utilizing the concept of illumination reflection model. This improved technique is based on Contrast Limited Adaptive Histogram Equalization (CLAHE) and reconstruction done using morphological processing with Top-hat transformation. The image is perceived in HSV color system and V component is estimated. This intensity (V) component is normalized and the inverse of the intensity component is computed. Then CLAHE algorithm is applied on the negative image. Then multiscale image enhancement is applied to the resultant image. Gamma enhancement is used to adaptively adjust the brightness component of an image. After gamma enhancement, the results of two gamma enhanced images with different gamma values are obtained. Principal Component Analysis extracts the significant information from these images that can be used for image fusion. During the PCA based image fusion, the weight value is adaptively calculated by using morphological Top-hat transformation to improve the quality of the image and illuminate the inconsistent background pixels. The proposed method focuses detail enhancement of extremely dark images by preserving the edges and structures. Experimental validations and results show the better performance of the proposed method than the existing method in terms of qualitative and quantitative measures.

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1. Introduction

We can grasp most significant information through Visual data or visual images. They correspond to the most significant amount of processed information captured or perceived which can be realized by the human brain. Image processing systems are widely used in several applications such as intelligent transportation, moving object tracking, biomedical imaging and driver assistance systems. The images acquired are blemished due to numerous defects that arise from a range of uncontrollable conditions like indoor lighting, nighttime lighting, cloudy weather. Under these conditions, the reflection of the tangent surface is weak and color is destroyed due to the presence of noise. Hence it results in serious impact of the image quality by causing content blur and detail loss in dark regions [5]. Low illumination images include compressed dynamic variation of the scene that can be enhanced for revealing hidden information. The proposed method results the dark pixel value of images are brighter as compared to the original low illumination images. Dark image enhancement or weakly illumination image enhancement is one of the challenging research areas which have applications in night vision, forensic applications, and under water enhancement. The extremely dark images contain uneven illumination of brightness value, which significantly degrade the image quality. The other challenges include addressing the noise during the enhancement of intensity component. The proposed method is aimed at enhancing and improving the vision of the extremely low illumination and dark images.

The research paper is put together as follows. The second section briefly introduces the literature and existing work related to low and weak illumination images. The third section presents the key steps of the method proposed in this paper including detailed explanation of the modules associated with the work. In section 4, the experimental results are thoroughly discussed and analyzed in qualitative and quantitative manner. The conclusion is summarized and presented in section 5.

2. Related Work

Different approaches for low illumination enhancement include methods based on histogram equalization [2], noise removal using wiener filter [4], linear contrast adjustment, CLAHE method and other existing methods. The author Veluchamy et al. [2] proposed a work based on image contrast, color enhancement using adaptive gamma correction and histogram equalization. Histogram equalization is used to enhance the digital images, but most cases it result in over illumination and intensity saturation. The work is based on performing gamma correction adaptively and histogram distribution method for contrast improvement. The gamma correction can be performed adaptively to adjust and enhance the contrast. The weighted histogram distribution presented in the work preserves the color and provides the fine details. Yuen Peng Loh et al. [3] discusses a CNN based work on dark and weakly illuminated image enhancement in which the Gaussian feature are used. The authors also introduced a new model for improving the less illuminated images as a localized function based on Gaussian smoothing. The Convolution Neural Network is learns the luminance distribution of dark images to understand the relation between the pixels. The main drawback of this method is over result which occurs due to the noise issues that arise from camera sensors when capturing low-light images. Xiankun Sun et.al [4] discussed a work on low light enhancement based on Guided Image Filtering in Gradient Domain. This technique is based on the model of illumination reflection and it is used to obtain the illumination component of the image using guided image filter. This method achieves impressive edge preservation and details results utilizing edge aware constraints in the gradient domain.

3. Proposed Work

An improved method for boosting or enhancing the extremely dark images based on histogram equalization i.e, CLAHE and structural reconstruction with Top-hat transformation is proposed. The process is shown in fig. 1

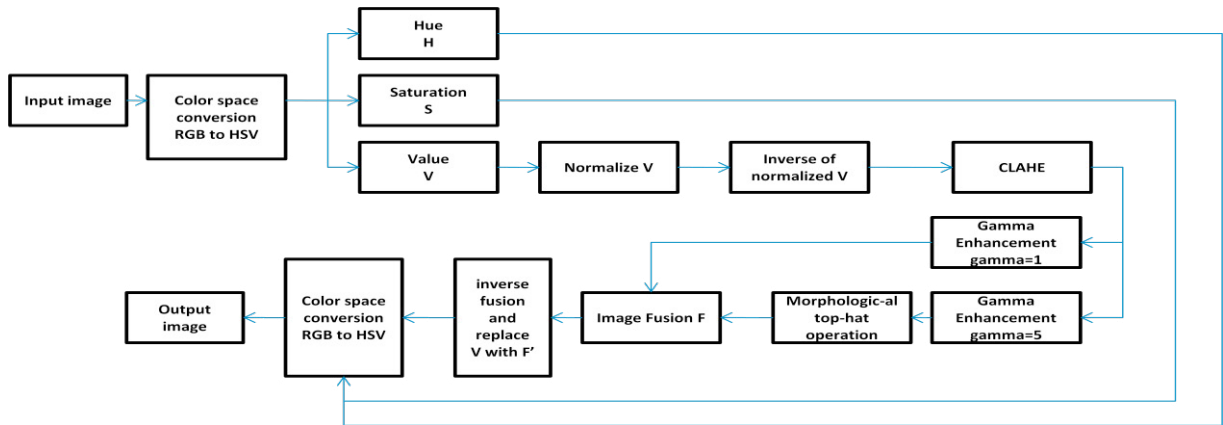


Fig. 1. Outline of the proposed method.

3.1. Conversion from RGB to Hue-Saturation-Value color model

Human eye has the capability of perceiving brightness rather than color. The illuminance correction is performed to retain better results. The illuminance correction must be performed on RGB channels directly; it is difficult to ensure that all channels are enhanced in a correct ratio, which often leads to distortion of the image color after the correction process. When an image is converted into HSV model, all the hue, saturation and intensity components are independent. So manipulation of the intensity values does not have any direct impact on the color of the images [1]. This is the reason the image in RGB color model is converted and transformed into the HSV color model. The hue $I_h(x, y)$, saturation $I_s(x, y)$, value $I_v(x, y)$ components of the image is computed and displayed in fig. 2.

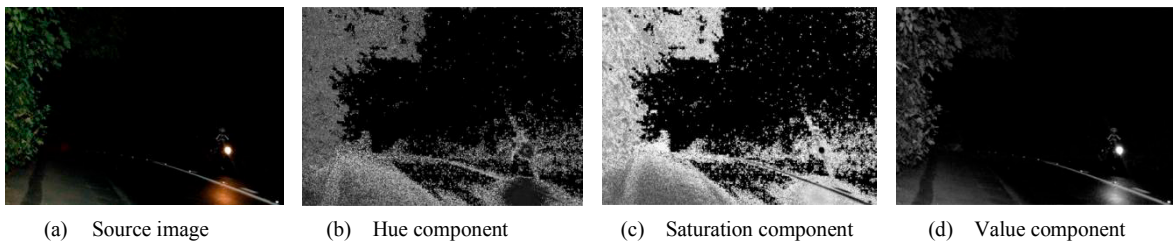


Fig. 2. Conversion from RGB space to HSV space.

3.2. Enhancement/ Improvement of illuminance component

3.2.1 Perform Normalization in intensity component

Normalization is a procedure that changes the pixel intensity values. Images can be represented as double precision numbers (or single precision) in the range 0 to 1. In the proposed work, the value of intensity component ranges between 0 and 255. Hence range the value within the interval [0, 1].

3.2.2 Inverse of the normalized intensity component

When intensity component is enhanced, large pixel noise will be introduced. In order to avoid noise enhancement of dark images, we take the inverse of the intensity component i.e.

$$I = 1 - V \quad (1)$$

3.2.3 Contrast Limited Adaptive Histogram Equalization (CLAHE)

CLAHE [10], a histogram based image enhancement method limits amplification based on the clipping done in the histogram to a limit it to a predefined level. In the proposed work, CLAHE method is used to enhance the inverse of intensity image. When CLAHE is applied in negative image, the visual features of image are improved and more details will be present in the enhanced image. The resultant image after applying contrasts limited adaptive histogram equalization in to the inverse of the normalized intensity component as shown in fig. 3.



Fig. 3. Example of CLAHE applied to the inverse of intensity image.

3.3. Gamma Enhancement

The illumination value of the over illuminated areas will decrease and illumination value of under illuminated area will increase as shown in Fig 4 and Fig 5. . To obtain an enhanced result, the gamma value has been experimentally found out as 1 and 5 and based on these values, gamma enhancement is done. The result of gamma enhancement with gamma value 1 returns the normal image. By increasing the value of the gamma output image appears as more enhanced or brighter. For the convenience here we use gamma value as 5. Gamma indicates the shape of the curve that describes the relationship between the input and output images. The gamma enhancement is mathematically expressed by using power law equation.

$$V = U^\gamma \quad (2)$$

where V is the enhanced output image, U indicates the input image and the value γ is denoted as gamma value.

3.4. Morphological Top-hat transform

Extremely dark image with uneven distribution of intensity values are effectively rectified by using morphological Top-hat transformation. i.e. the result of gamma enhancement with gamma value 1 return the normal image while the result of gamma enhancement with gamma value 5 return the darker image. The fused output image is an image with uneven illumination components. To rectify it the proposed method introduce an approach known as morphological Top-hat transform, that can be used to enhance the contrast in a low illumination images by adaptively calculating the weight values in PCA fusion. Top-hat transform [9] is performed by using a disk structuring element with radius 3. This resultant value is used to compute the weight value during the PCA fusion.



CLAHE applied to the inverse image Enhanced image with gamma value 1 Enhanced image with gamma value 5

Fig. 4. CLAHE enhancement and gamma enhanced images with multiscale gamma values.

3.5 Image fusion

The fused images just calculate the weighted sum of the source image.i.e.

$$F = \sum_{i=1}^N \omega_i S_i \quad (3)$$

where F - fused result image, ω_i - weight coefficient and S_i - image to be fused. Image fusion is done with Principal

Component Analysis (PCA) algorithm [1] is used to determine weighting coefficients.

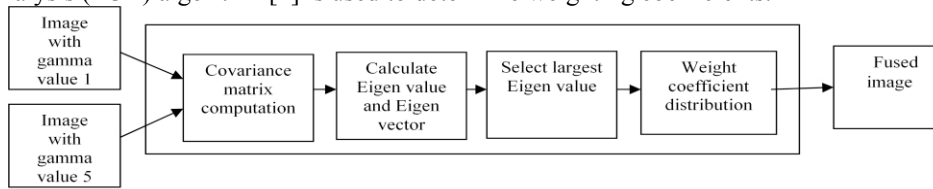


Fig. 5. Image fusion using PCA.

The steps of the image fusion are as implemented from [1]. In the proposed method, the output of gamma enhanced image with gamma value one and output of gamma enhanced image with gamma value five are fused together to get the result.

3.5. Inverse image fusion

As shown in Fig 6 and Fig 7, after the fused image is obtained it is subtracted from one to get the inverse value of fused image i.e.

$$F' = 1 - F \quad (4)$$

After performing inverse fusion, the value of V component is replaced into F' the value of inverse fusion.

$$V = F' \quad (5)$$



Fig. 6. Image fusion based on PCA.



Fig. 7. Inverse of the fused image.

3.6. HSV to RGB color space conversion

The HSV image is converted back to RGB space using formula [6].

The summary of the proposed dark image enhancement algorithm is given as follows.

Proposed Algorithm

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- Step 1: Load dark image I as input.
 - Step 2: Convert the image from RGB space to HSV color system.
 - Step 3: Normalize the intensity component and take the inverse of normalized intensity component .
 - Step 4: Perform the Contrast Limited Adaptive Histogram Equalization (CLAHE) into the resultant image obtained from step 3.
 - Step 5: Let $\gamma = 1$ and $\gamma = 5$ and obtain the gamma enhanced images.
 - Step 6: Calculate the weight values with morphological Top-hat transform.
 - Step 7: Obtain the resultant image after fusion.
 - Step 8: Obtain the inverse of the fused image.
 - Step 9: Use F' as the V component to merge H and S component from HSV and convert image J into RGB.
 - Step 10: Output the enhanced color image J.
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4. Results and Discussion

In order to verify the performance of the proposed work, we tested our method in extremely dark night-time images, indoor images and so on.

4.1 Qualitative Analysis

Experimental results on various images show that proposed method has an obvious superiority in preserving detail and can enhance the image quality especially in case of extremely dark images. Some of the experimental results are shown in fig. 8. The Exdark dataset is used for the experimental analysis. The dataset available in github <https://github.com/cs-chan/Exclusively-Dark-Image-Dataset>.

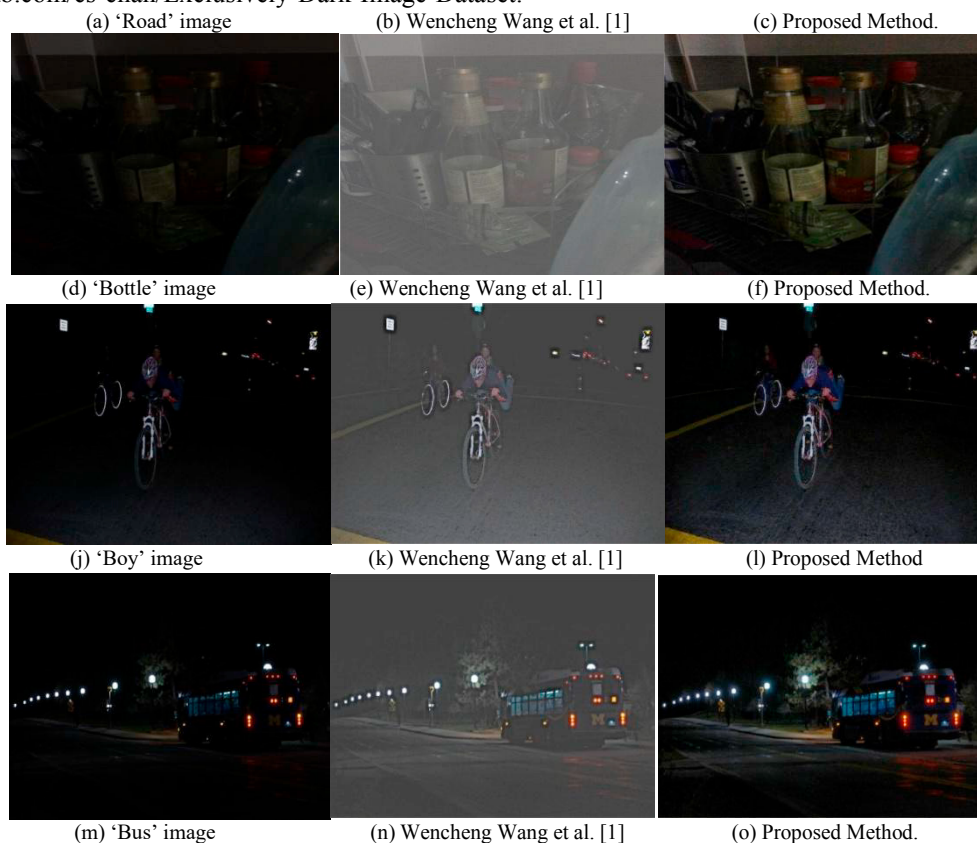


Fig. 8. Enhanced images with extremely low illumination.

4.2 Performance metrics

In order to estimate the efficiency of this method, three approaches are used i.e. BRISQUE [7], PIQE and NIQE [8]. These techniques are used to assess the performances of low illuminated images by comparing the input image and the enhanced image. The lowest values indicate the higher perception quality of the images. The table 1 shows the average quantitative performance of ExDark dataset. The work compare the quantitative analysis with Wencheng Wang [1]. The proposed method shows better quality perception than Wencheng Wang [1].

Table 1. Average Quantitative Performance based on *NIQE*, *PIQE* and *BRISQUE*.

ExDark dataset	Proposed Work	Wencheng Wang et al.[1]
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	Niqe	Piqe	Brisque	Niqe	Piqe	Brisque
Bicycle	<u>3.7758</u>	<u>37.8515</u>	<u>26.8796</u>	9.4255	98.8730	46.8923
Boat	<u>3.4714</u>	<u>38.2349</u>	<u>24.8571</u>	3.8800	99.2705	27.4133
Bottle	<u>3.9224</u>	<u>37.1810</u>	<u>28.8448</u>	9.9899	98.9763	46.4096
Bus	<u>3.6729</u>	<u>36.1322</u>	<u>26.5876</u>	3.7698	99.4329	26.9319
Car	<u>3.4314</u>	<u>35.6315</u>	<u>29.1532</u>	9.9528	99.4420	46.7260
Cat	<u>4.0760</u>	<u>35.6371</u>	<u>31.5174</u>	4.1683	99.0578	34.0022
Chair	<u>3.7832</u>	<u>38.7956</u>	<u>29.0748</u>	3.9815	98.9283	31.3371
Cup	<u>3.5799</u>	<u>37.4407</u>	<u>29.5414</u>	3.9446	98.4408	33.1276
Dog	<u>4.0097</u>	<u>37.5215</u>	<u>30.0219</u>	9.9864	99.1261	46.2956
Motorbike	<u>3.8571</u>	<u>38.1524</u>	<u>28.0163</u>	9.5071	99.0029	46.8439
People	<u>3.9921</u>	<u>30.0680</u>	<u>28.8918</u>	9.8564	99.0072	46.3867
Table	<u>3.5883</u>	<u>37.7191</u>	<u>29.0168</u>	9.8328	98.5334	46.7486

5. Conclusion

In this work, we introduce a dark image enhancement based on Contrast Limited Adaptive Histogram Equalization technique and Morphological operations followed by fusion. The RGB image is converted into HSV color space, from that image intensity component is extracted and it is normalized. Intensity component contain extremely dark pixels and when intensity value is enhanced, pixel noises will be introduced. So we take the inverse of the intensity component before removing the noise. The Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to the inverse of the intensity component. It will enhance the image. Then gamma enhancement with multiscale gamma value is performed. Image fusion is done by PCA. During image fusion, calculate the weight value adaptively by using morphological Top-hat transformation to improve the quality of the image. Inconsistent background illumination can be adjusted using top-hat transform. Next, the inverse of image fusion is computed. In the last step, the image in HSV color model is converted and transformed back to the RGB color model. Compared with Wang et al., the experiment results demonstrate that our method can achieve more edge preservation, detailed enhancement and balances the color across the images. This method can be efficient for real time extremely dark image enhancement applications.

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